A Named Entity Recognition Pipeline for Custom Anonymisation of Clinical Free-text Notes

Andreia Rogerio[1], Chris Boustred[1], Nicholas Broad[2]*, Niels Rogge[2]*, Florent Gbelidji[2]*, Anna Need[1], Francisco Azuaje[1]

[1]Genomics England Limited, [2]HuggingFace * Contributed Equally

Abstract

Sharing clinical data leads to faster collaboration in medical research and ultimately contributes to disease prevention and improved treatments. At Genomics England, data sharing is crucial to research and to enable the infinity loop that allows continuous improvement of cancer and rare disease diagnosis.

Unfortunately, ensuring personal identifiable data (PID) is anonymized and compliant with the relevant governance is still challenging, especially in unstructured data. In the Clinical Variant Ark (CVA) Portal, clinical scientist share their interpretations of genomic variants in free text boxes. This is highly valuable information but can only be shared with researchers if completely anonymised.

Here we propose a prof of concept solution to identify and redact all personal names in the CVA database, while still maintaining the relevant medical content. We show that with NLP techniques the anonymisation can reach values of 100% of recall and 80% of precision.

Methodology and Results

We applied natural language processing (NLP) pre-trained models for named entity recognition (NER) to detect person names in the CVA entries. **Upon manual review and labelling**, 210 records were found to contain PID, which together consist of the evaluation set:

- 95 records with PID in the genomic interpretation (GI) field
- 115 records with PID in variant comments (VC) field

We compare the recall obtained by **different open-source pre-trained NER models**, available in HuggingFace, and integrated in the anonymisation pipeline tool Presidio. We further add NLP techniques to improve the precision of our solution. The most performant model was Flair with 100% and 85% recall for the VC and GI subsets respectively.

Technical details

To find the best pre-trained model we took a sequential approach:

1. We first tested several NER models and selected the most performant based on **maximum recall** obtained when compared with the evaluation set. (Figure 1)
   a) Application of each model to each entry
   b) Storing all predicted PID for a given entry-model pair and scanning a second time (to find the same entries in different places of the text)
   c) Generate case-variations of all predicted PID for a given entry-model pair and scanning a third time.
   d) Cleaning manual labelling of false positives.

2. We then took the most performant model and applied further NLP techniques to **improve its precision** (Fig. 2)
   a) we developed a RegEx pattern-matching recogniser that detects literature authors fully integrated in Presidio;
   b) we included the pre-trained model **ner-disease-ncbi-bionlp-bc5cdr-pubmed** to recognise disease names and remove them from final predictions.

For the GI dataset the best performance was obtained using the same strategy and reached 85% of recall and 67% of precision.

Figure 1: Maximising Recall for VC subset

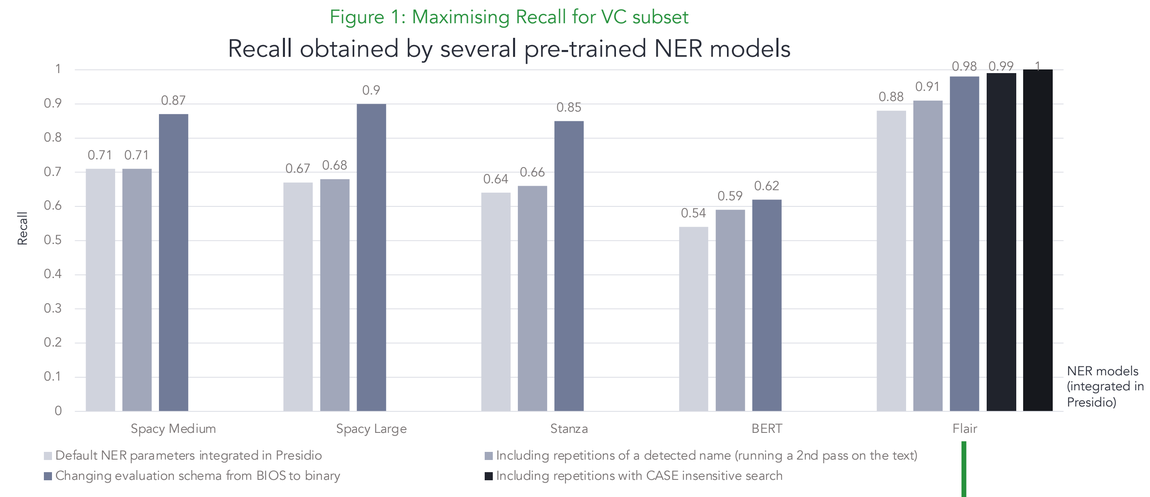
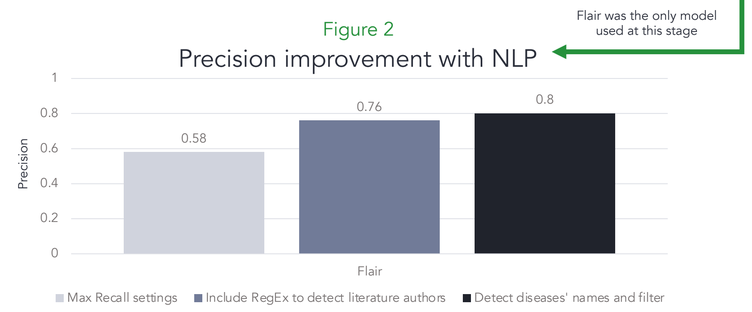


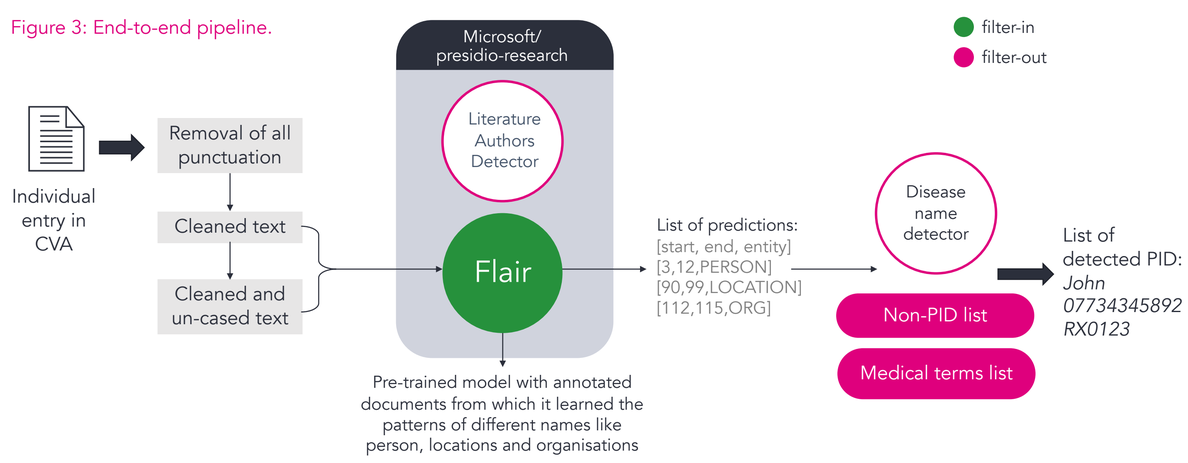
Figure 2



Final Remarks

- Using pre-trained models as baseline approach, and improving them with NLP techniques, proved to be an efficient way to build a customised solution.
- Flair outperforms other better known models, which is related with the underlying structure of the model: Flair is a BiLSTM-CRF character based monolingual model pretrained on Wikipedia, making it very good at NER because suffixes and prefixes are powerful features.
- Another factor that greatly improve the performance metrics was the manual review, i.e. investigating the cases where the model underperforms (names that are not detect). The performance difference between upper-cased words and lower-cased words is an example of the benefit of manually reviewing the results.
- These results show high potential for Flair applicability in other unstructured clinical datasets, while reinforcing the need for a human-in-the-loop to refine the results' quality.

Figure 3: End-to-end pipeline.



Acknowledgements and References

Web sources:

Flair GitHub repository: zalandoresearch/flair (accessed August 2022), HuggingFace Flair model source: flair/ner-english-large (accessed August 2022), Microsoft Presidio GitHub Repository: microsoft/presidio (accessed August 2022), Mixed Case limitation: StandfordNLP website, TrueCase tool - https://stanfordnlp.github.io/CoreNLP/truecase.html (accessed August 2022)

Literature Articles:

Hasni, S., Faiz, S. Word embeddings and deep learning for location prediction: tracking Coronavirus from British and American tweets. *Soc. Netw. Anal. Min.* **11**, 66 (2021). https://doi.org/10.1007/s13278-021-00777-5

Alan Akbik, Tanja Bergmann, Duncan Blythe, Kashif Rasul, Stefan Schweter, and Roland Vollgraf. 2019. FLAIR: An Easy-to-Use Framework for State-of-the-Art NLP. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (Demonstrations)*, pages 54–59, Minneapolis, Minnesota. Association for Computational Linguistics.

Contact: Andreia Rogério | Applied Machine Learning Engineer | andreia.rogerio@genomicsengland.co.uk